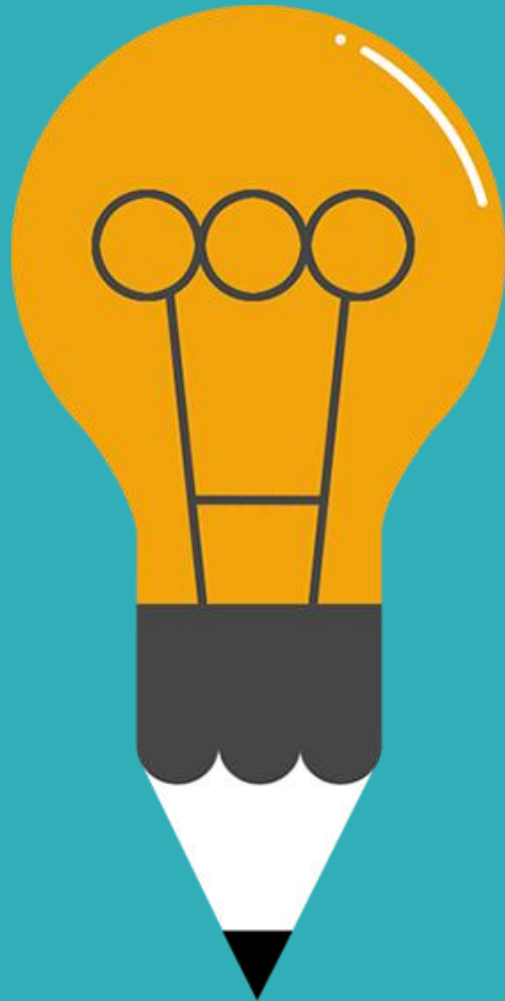




DATA SCIENCE PROJECT

Project Description



01

Project

Business Intelligence

02

Primary Topic

DPV

03

Dataset

Sakila DB

Rental Business Process

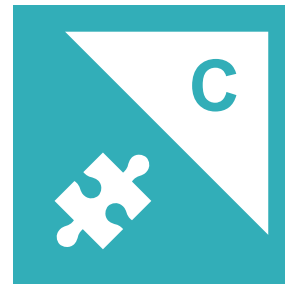
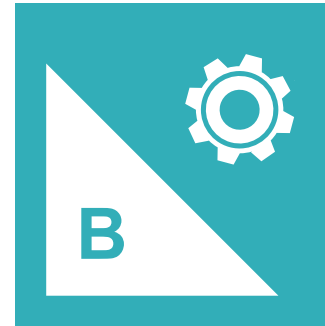
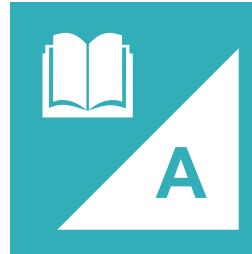
- Customer visits the store, Picks a DVD
- Customer paid the rent
- DVD took out of inventory by the staff member
- Customer returns home and enjoys DVD
- Customer returns to store with DVD
- DVD returned to the staff member

Business Question

Balance Scorecard Perspective

Revenue Perspective

How to optimize pricing strategies for DVD rental business to maximize profitability?



Customer Perspective

What are the customer demographics and preferences that are most likely to rent a higher number of DVDs and have a higher likelihood of returning as repeat customers?

Learning and Growth Perspective

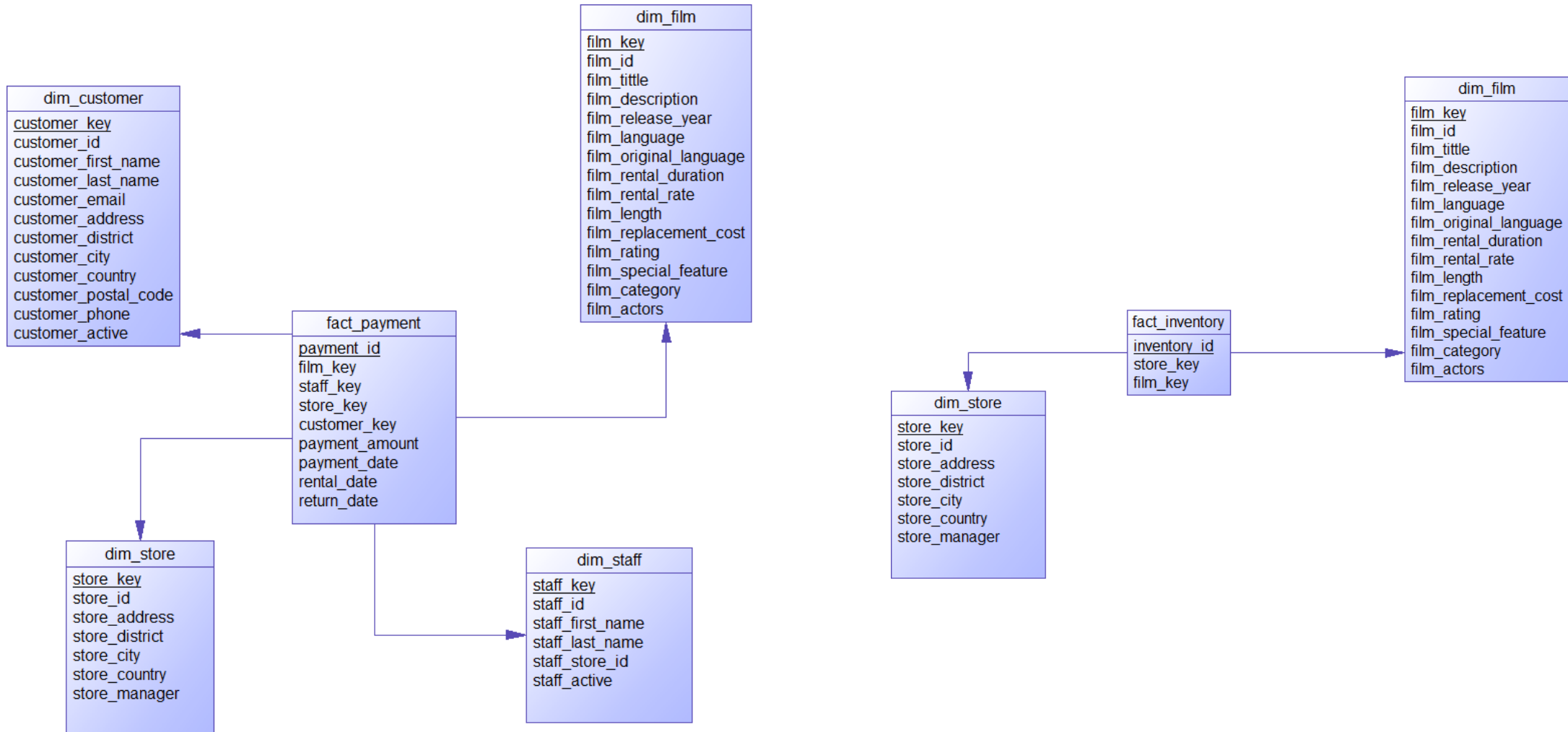
What is the impact of staff experience levels on rental return rates?

Internal Process Perspective

What are the internal processes and factors that have the most significant impact on the efficiency of DVD inventory?

Star Schema

Insert the title of your subtitle Here



ETL

- Connect to the source database: Establish a connection to the source database MySQL using MySQL Connector.
- Extract data: Use SQL queries to extract data from the source database tables. Store the extracted data frame Pandas data frame.
- Clean and transform data: Clean and change the extracted data using Pandas functions and methods.
- Create tables in the data warehouse as defined in the star schema
- Load data into the data warehouse: Connect to the PostgreSQL data warehouse using a Python library psycopg2. Then load data to the data warehouse in `bronto.ewi.utwente.nl`



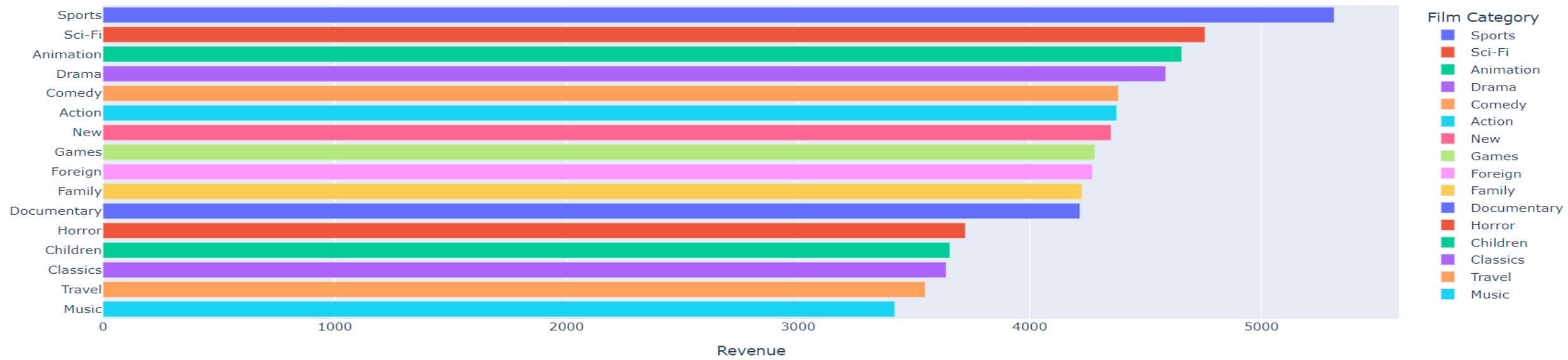
Visualization

Created using Python

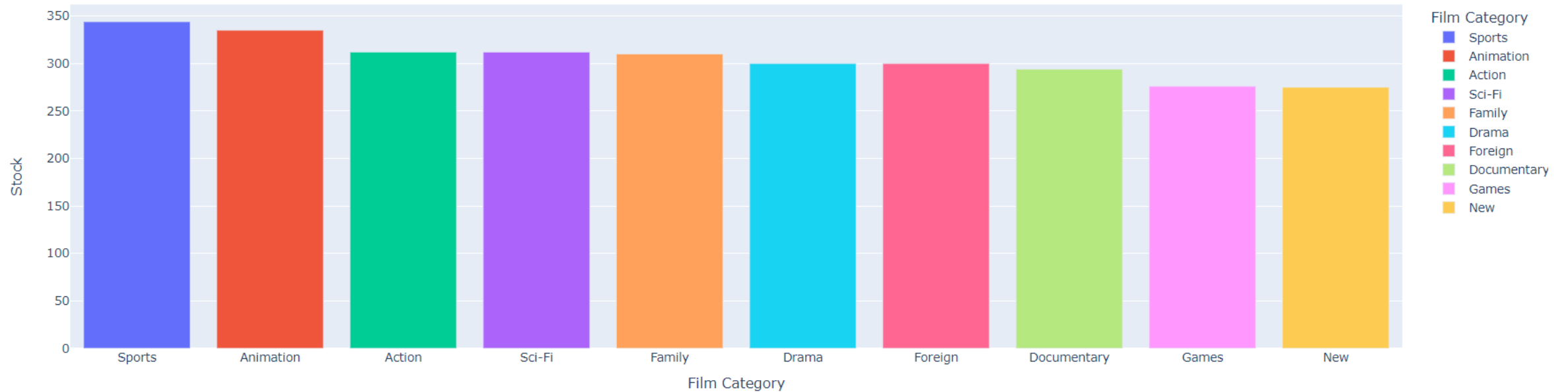
Financial Perspective

- Revenue
- Revenue per customer
- Rental price per movie
- Revenue per category film
- Popularity of films
- What is the most feature concerned by the customer?

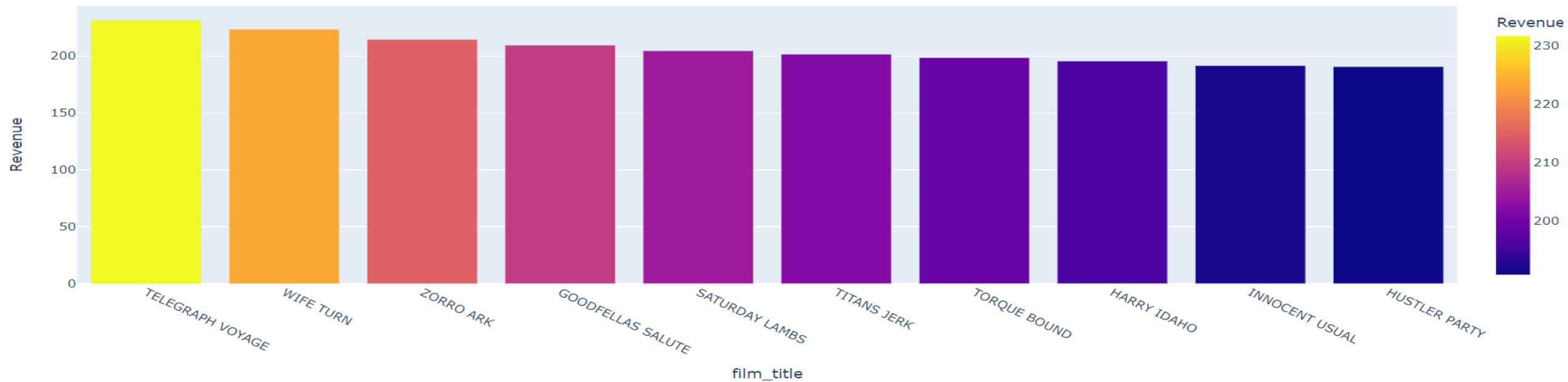
Revenue per Category



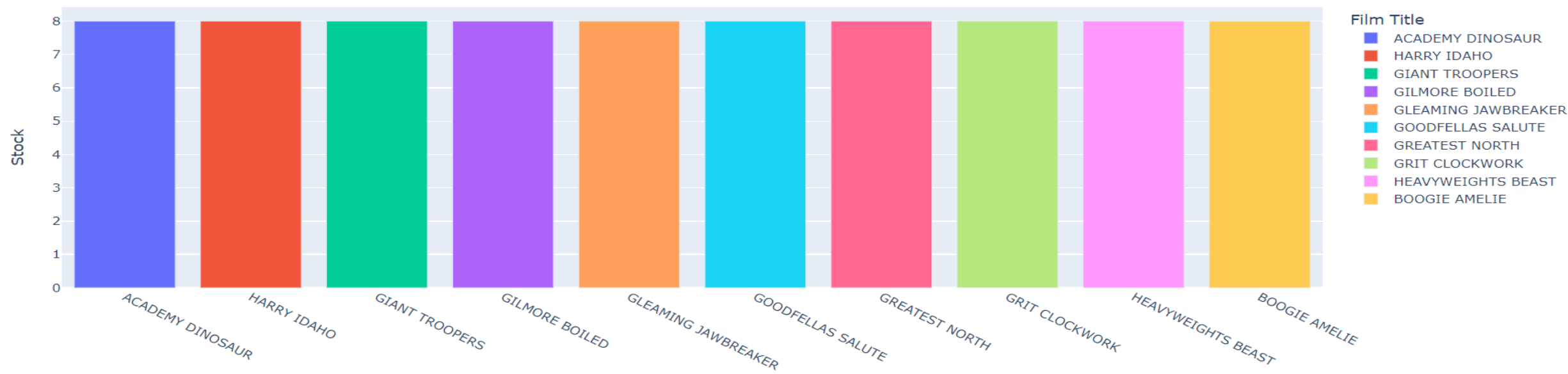
Stock per Category(Top 10)



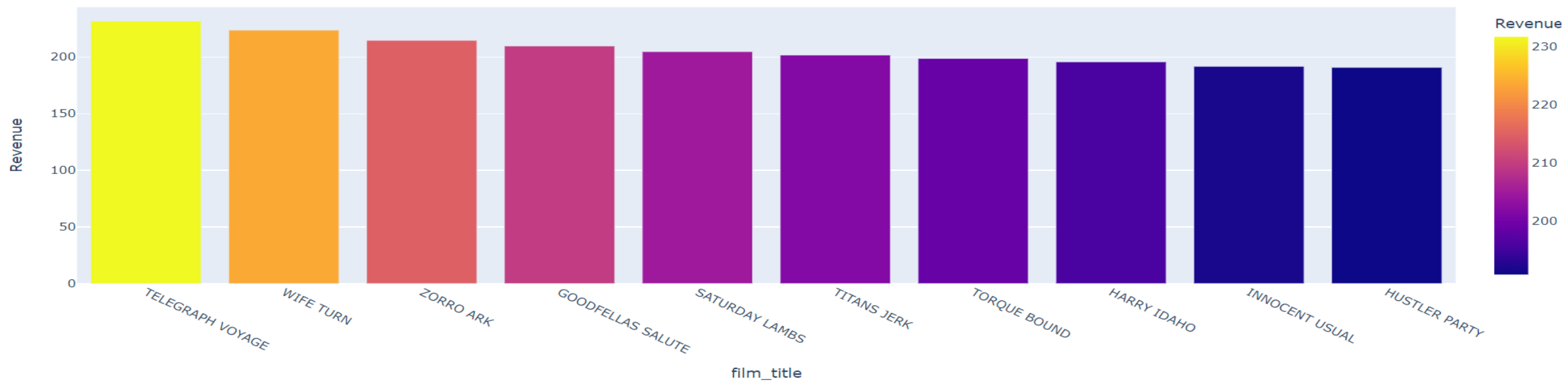
Revenue per Movie (Top 10)



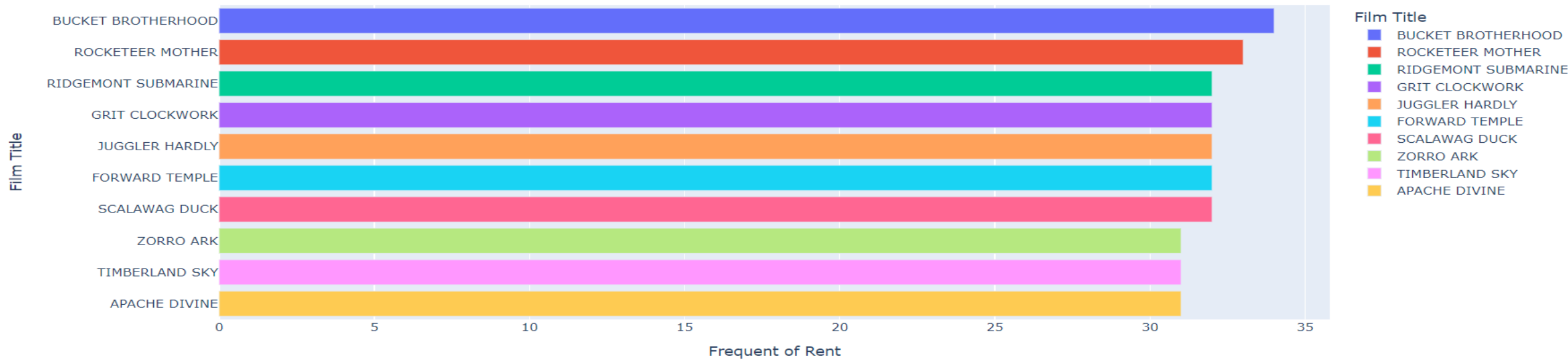
Stock per Movie(Top 10)



Revenue per Movie (Top 10)



frequent rental of movie

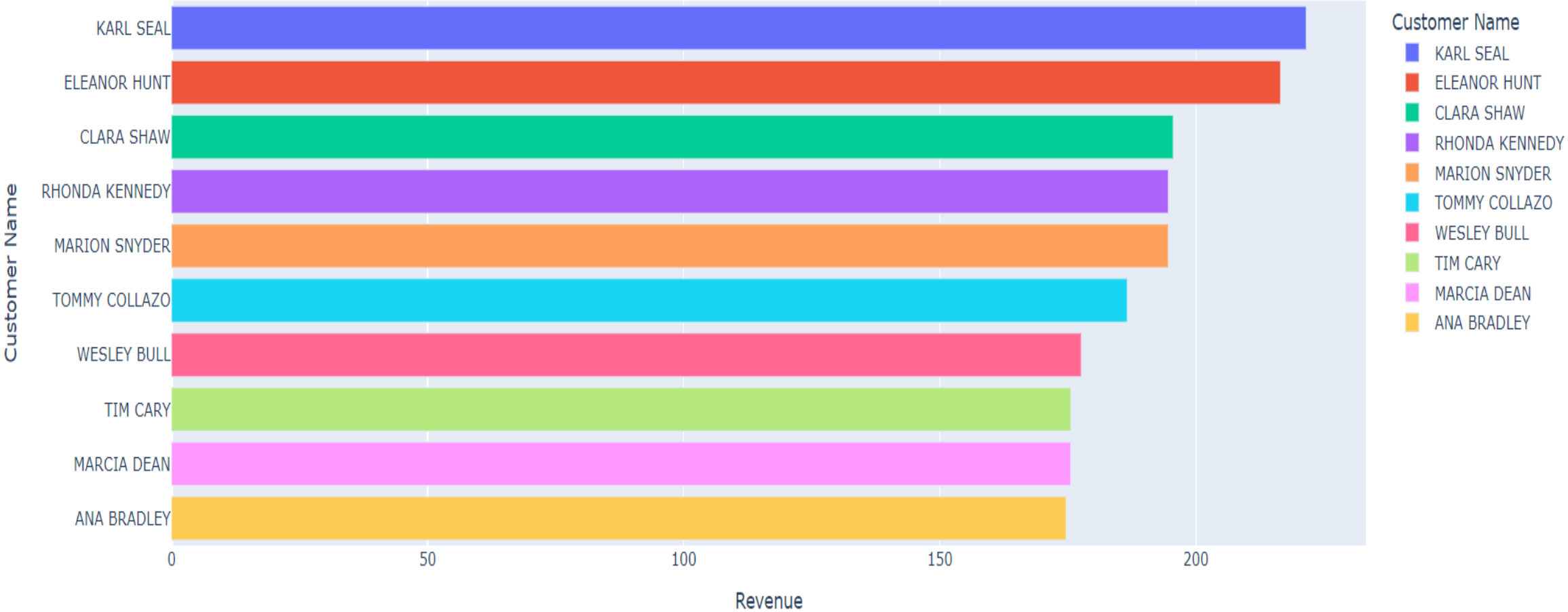


film_special_feature	payment_amount
Trailers,Commentaries,Behind the Scenes	5517.78
Commentaries	5017.99
Trailers,Commentaries	4827.59
Behind the Scenes,Commentaries,Deleted Scenes	4806.89
Trailers	4753.49
Trailers,Behind the Scenes	4676.64
Behind the Scenes	4655.87
Commentaries,Behind the Scenes	4615.03
Commentaries,Deleted Scenes	4600.78
Trailers,Deleted Scenes	4369.36

Customer Perspective

- Most valued Customer
- RFM (Recency, Frequency, Monetary) Analysis

Most Value Customer (Top 10)

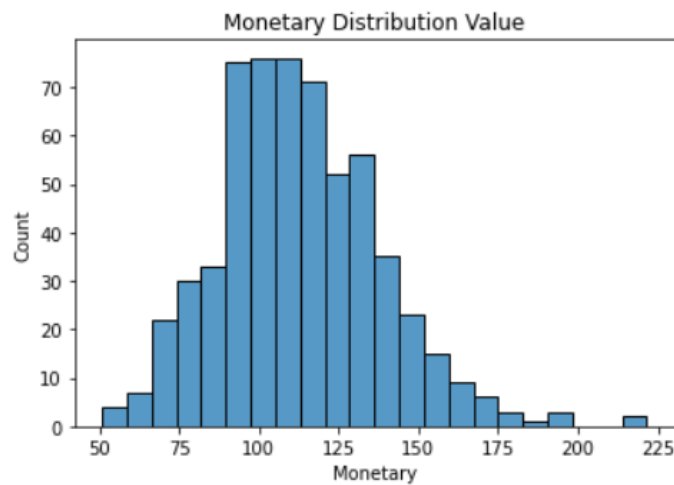
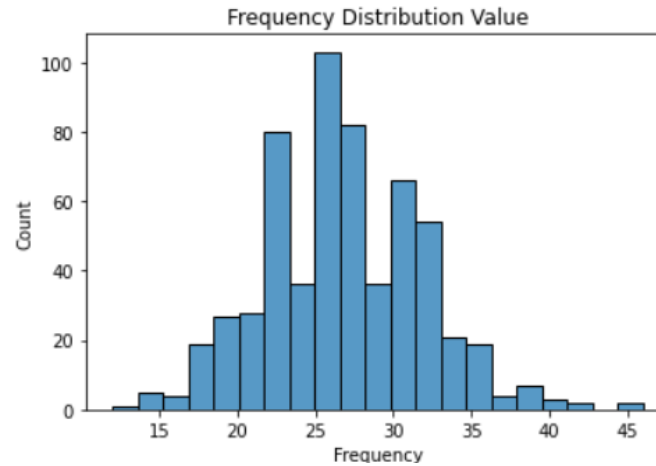
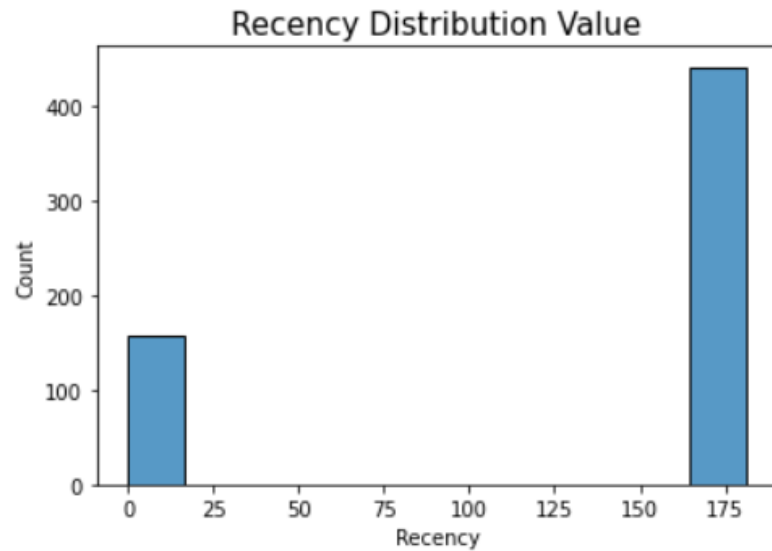


RFM

(Recency, Frequency, Monetary)

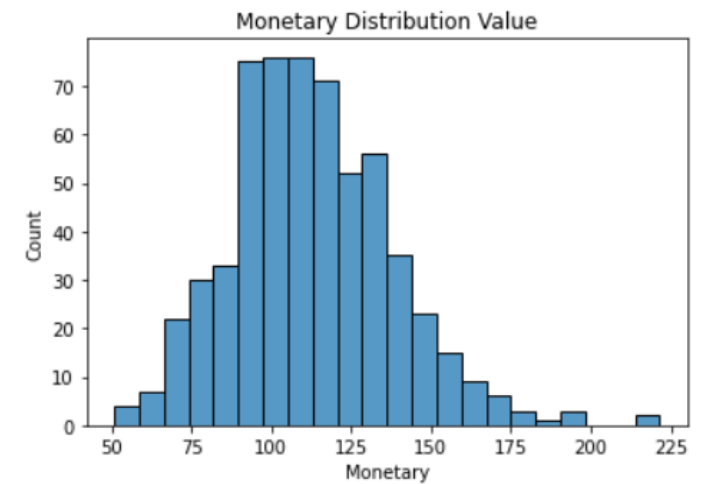
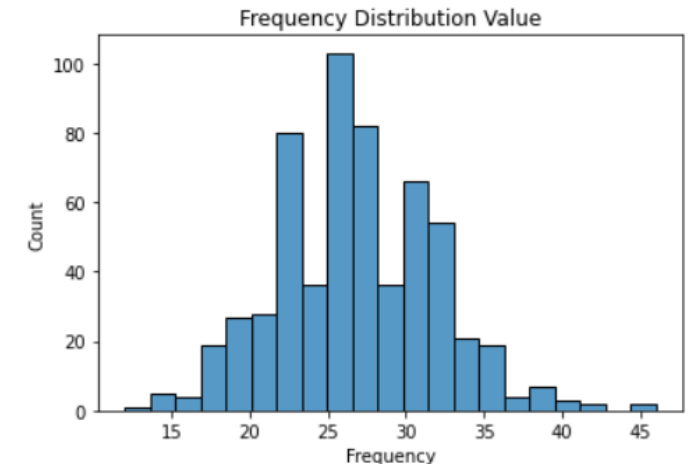
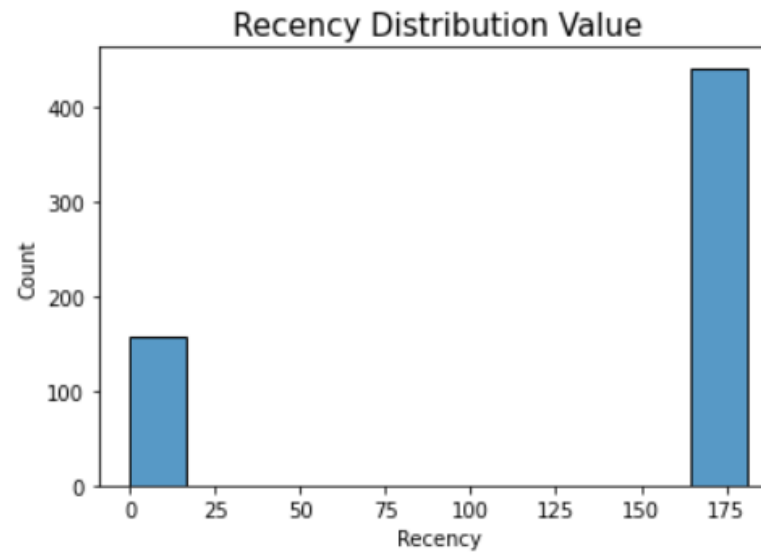
Analysis

	customer_key	Recency	Frequency	Monetary
0	1	176	32	118.68
1	2	175	27	128.73
2	3	175	26	135.74
3	4	175	22	81.78
4	5	0	38	144.62



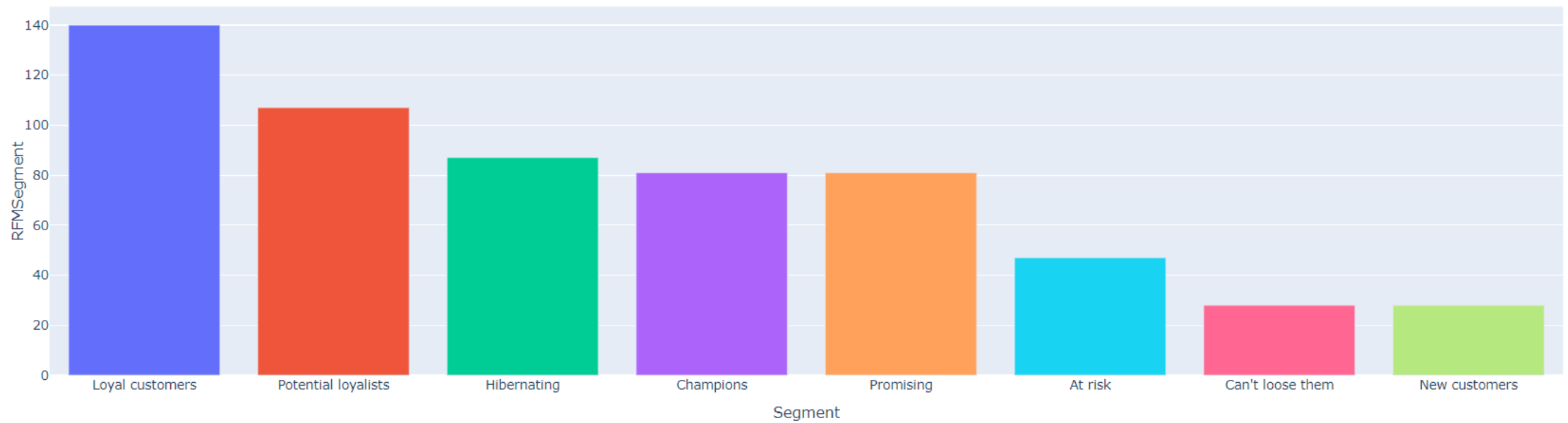
RFM (Recency, Frequency, and Monetary) Analysis

	customer_key	Recency	Frequency	Monetary
0	1	176	32	118.68
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3	4	175	22	81.78
4	5	0	38	144.62



	customer_key	Recency	Frequency	Monetary	R	F	M	RFMScore	RFMSegment	Segment
0	1	176	32	118.68	2	4	3	9	243	Can't loose them
1	2	175	27	128.73	3	3	4	10	334	Loyal customers
2	3	175	26	135.74	3	2	4	9	324	Potential loyalists
3	4	175	22	81.78	3	1	1	5	311	Promising
4	5	0	38	144.62	4	4	4	12	444	Champions
5	6	175	28	93.72	3	3	1	7	331	Loyal customers
6	7	177	33	151.67	1	4	4	9	144	Can't loose them
7	8	175	24	92.76	3	2	1	6	321	Potential loyalists
8	9	0	23	89.77	4	1	1	6	411	New customers
9	10	176	25	99.75	2	2	2	6	222	Hibernating

Customer Segmentation Distribution

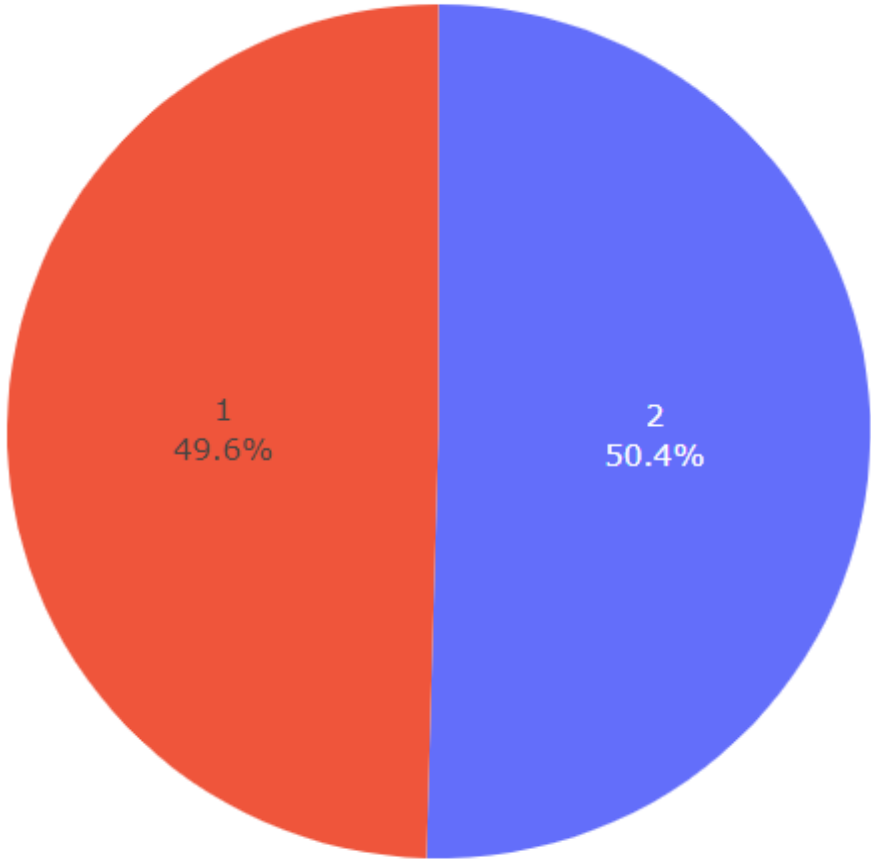


- **Hibernating:** Rebranding value. Offer relevant products and give good offers
- **Champions:** Give Loyalty Rewards or give chance for them to try a new product
- **Loyal Customer:** Sell higher value products and ask for their review to know more about their experience
- **At Risk:** Send an email blast to reconnect to them and give a product recommendation
- **Potential Royalist:** Give product recommendations and loyalty rewards to engage them
- **Can't Lose Them:** Give special offers to them to make them feel valuable again
- **Promising:** Make brand awareness
- **New Customer:** Start building relationships by providing a good experience

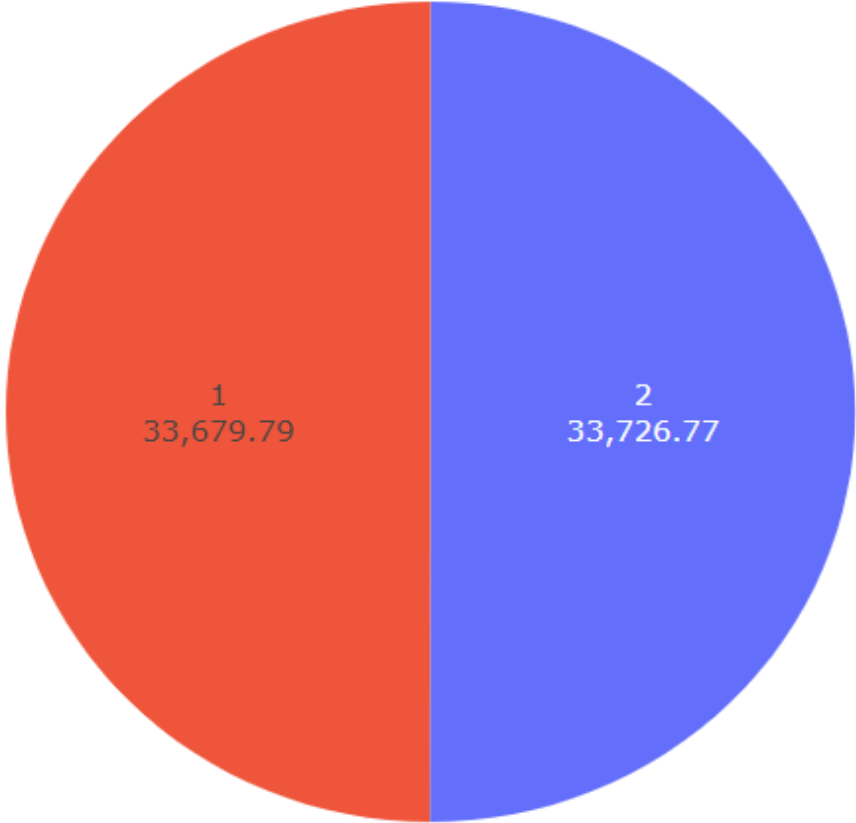
Internal Process Perspective

- Number of Inventory per store
- Total Revenue Per store
- Average duration rent per film compared to actual duration
- Duration of rent per film compare to film length
- Average duration rent per category film

Number Stock Per Store



Revenue Per Store



Minimal Rental Duration : 3 days
Maximum Rental Duration : 7 days

film_rental_duration

0

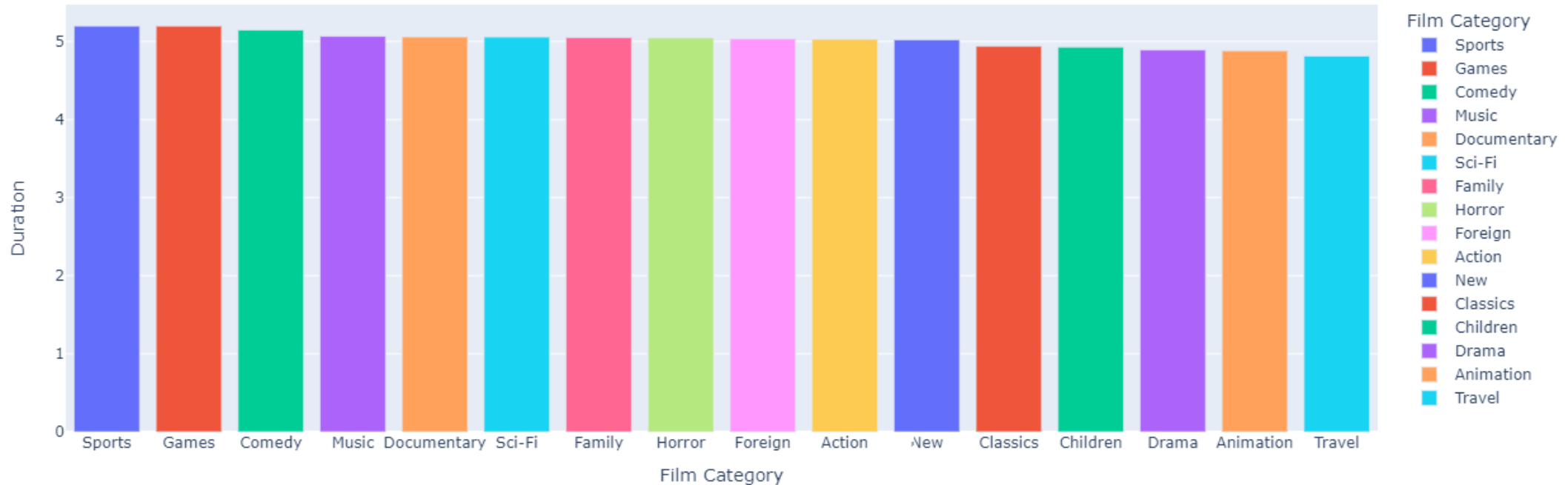
7

film_rental_duration

16043

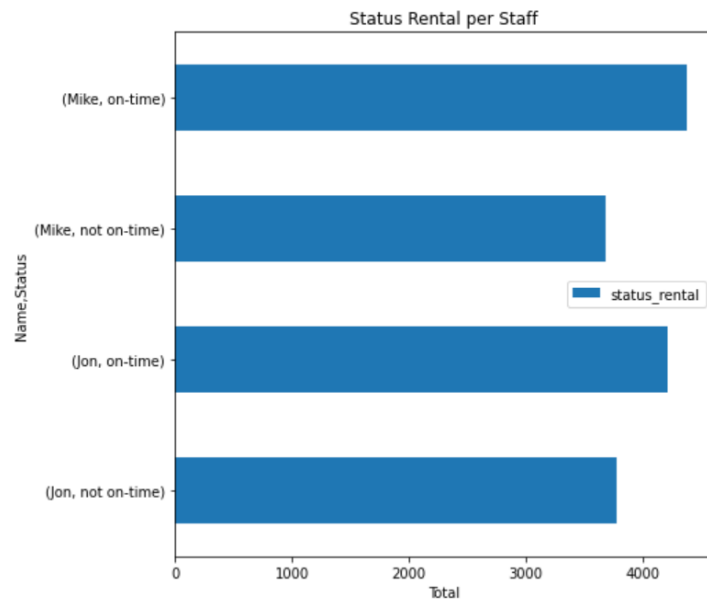
3

Average Duration Rental per Category

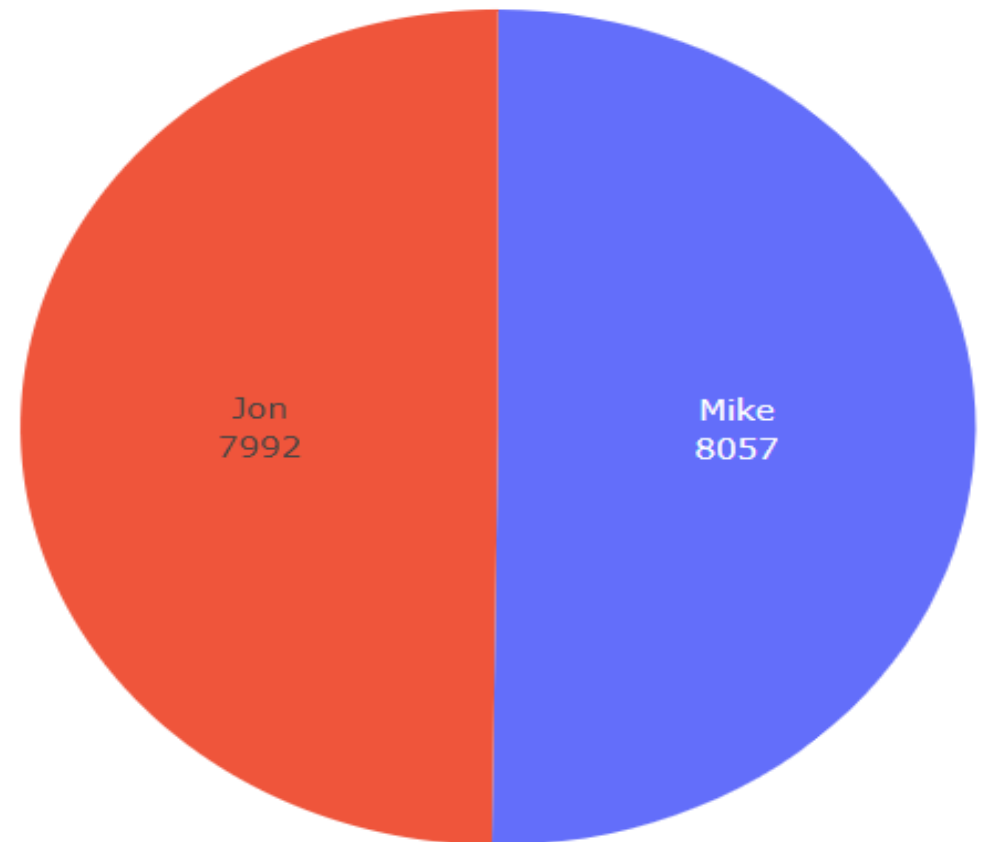


Innovation & Growth Perspective

- Staff experience levels: A metric that measures the experience of employees.
- Rental return rates: A metric that measures the percentage of DVDs returned on time by customers.



status_rental		
staff_first_name	status_rental	
Jon	not on-time	3774
	on-time	4216
Mike	not on-time	3678
	on-time	4376



Conclusion

- The Pricing strategy can be adjusted to maximize the profit
- Minimum rental duration can be changed to 4 days since the average rental duration is between 3 – 5 days
- Do the recommendation to maintain good relations with customers



Thank you

Insert the title of your subtitle Here